Bridging Evolution and Learning: Attention-Guided Co-Design of Modular Soft Robots

Author Name
Affiliation
email@example.com

Abstract

Million years of biological evolution has witnessed the co-evolution of creature's brain and organism to adapt to their surroundings. Likewise, modular soft robots that mimic life forms require brain-morphology co-design, while the vast morphology search space and the high computational cost of fitness evaluation make it challenging. Traditional approaches such as the genetic algorithm (GA) or Bayesian optimization (BO) typically converge slowly and incur substantial computational overhead. To address these issues, this paper proposes Attention-Guided Co-Design (AGCD), a co-design method guided by attention weights. Based on the theoretical mechanism of the Baldwin Effect, we employ the attention weights learned by control strategies to guide offspring structure generation, enabling offspring to more readily develop advantageous phenotypic traits of their parents, thereby achieving an effective bridge between evolution and learning. To accelerate this process, we introduce a multi-maturity evaluation and incorporate both maturity and fitness as ranking objectives in non-dominated sorting, thereby maintaining a balance between task performance and evaluation cost during evolution and avoiding expensive full evaluations of candidate individuals at early stages. Extensive experiments demonstrated that, under the same computational budget, our proposed method AGCD outperformed baseline methods in terms of search efficiency, final performance, and zero-shot generalization.

1 Introduction

Embodied intelligence emerges from the dynamic interaction between the brain, body, and environment [Liu *et al.*, 2025b; Pfeifer *et al.*, 2006]. Throughout evolutionary history, the evolution of natural organisms has been driven by the coupled interaction of neural control and physical form. Inspired by this phenomenon, Sims [1994] pioneered the co-evolution of robot morphology and controllers. This idea promoted the development of automated robot design by a joint optimization of the interdependent body structures and neural controllers at the design stage [Lipson and Pollack, 2000]. In this framework, robotic control strategies and physical structures are evolved jointly during co-design [Bongard and Pfeifer, 2003; Liu *et al.*, 2025a]. However, effective shared optimization of robotic morphology and the controller remains challenging due to the high dimensionality and strong coupling of the design space and the control policy space [Cheney *et al.*, 2016].

Robot morphology-control co-design can be formulated as a bi-level optimization problem. Due to the tight coupling between body structure and control, changes in morphology often invalidate previously effective control strategies leading to performance degradation [Cheney *et al.*, 2018]. In addition, the morphological design space is extremely large and requires extensive exploration, while complex interactions among robot components make the learning of effective control strategies hard [Zhao *et al.*, 2025a; Zhao *et al.*, 2024]. Existing co-design approaches can be broadly categorized into two paradigms: evolutionary computation-based methods and reinforcement learning-based methods. Evolutionary computation methods typically employ a parallel population-based search to jointly optimize morphology and controllers [Stanley *et al.*, 2009; Tanaka and Aranha, 2022; Harada and Iba, 2024]. However, fitness evaluation for each candidate morphology is expensive because it often requires training a near-optimal controller. Under limited computational budgets, this training process becomes time-consuming, making morphology-control co-design a computationally expensive optimization problem [Liu *et al.*, 2023]. Moreover, both methods can be prone to stagnation in suboptimal regions and may fail to effectively exploit fitness feedback to guide the search [Mertan and Cheney, 2025; Mertan and Cheney, 2024a]. Reinforcement learning-based approaches incorporate morphology design into a Markov Decision Process (MDP) [Wang *et al.*, 2023c; Wang *et al.*, 2019; Lu *et al.*, 2025] and use gradient information to guide the search [Xu *et al.*, 2021]. However, the high sensitivity to reward shaping can cause optimization to become trapped in local optima, particularly in tasks with sparse rewards [Saito *et al.*, 2022].

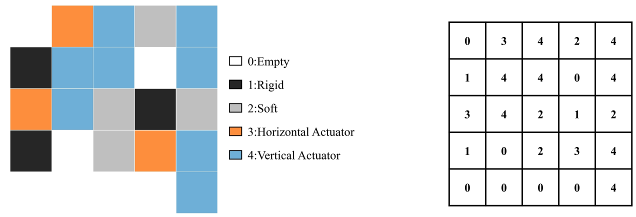
To address these issues, we propose Attention-Guided Co-Design (AGCD), a co-design approach that bridges evolution and learning through the attention weights. Within the theoretical framework of the Baldwin Effect, learned behaviors are not directly inherited by offspring. Instead, structures that make such behaviors easier to learn are more likely to appear in later generations. Since attention learned during controller training can reflect the contribution of each robot component to the task, we use attention weights to induce this structural bias. By guiding the generation of offspring structures, we increase the probability that offspring will develop a structural bias that facilitates the acquisition of important traits from their parents. To accelerate this process, we propose a multi-maturity evaluation strategy, in which low-quality individuals are filtered out at low maturity levels, ensuring that the costly evaluation budget is concentrated on high-potential individuals. During environmental selection, maturity and fitness are treated as non-dominated objectives, allowing both low-maturity individuals and high-potential individuals to be retained simultaneously. This design maintains a balance between task performance and evaluation cost throughout the evolutionary process. Furthermore, to obtain reliable attention weights that accurately reflect the contribution of each robotic component, we propose a self-attention-based control strategy. The main contributions of this study are as follows:

- We propose AGCD, an attention-guided co-design framework that follows the theoretical mechanism of the Baldwin effect, employing attention-based variation operator to improve offspring generation quality.
- We introduce multi-maturity evaluation, employing Maturity-Fitness Non-Dominated Sorting to protect immature yet promising individuals and to boost overall evolutionary efficiency.
- We propose a self-attention control strategy and utilize coordinate-based positional embedding (CoordPE) to enhance positional awareness, bringing efficient control decisions.

2 Related Work

2.1 Robot Co-design

Robot co-design aims to jointly optimize morphology and control strategies. Due to the close interdependence between morphology and the controllers [Rosendo *et al.*, 2017; Ma *et al.*, 2023], morphological design often directly influences the expressive capacity of control strategies. Previous studies have examined how different encoding approaches affect the co-optimization performance [Pigozzi *et al.*, 2023; Bhatia *et al.*, 2021; Stanley, 2007], and these methods are commonly categorized as direct encoding or indirect encoding. Among indirect methods, the compositional pattern-producing network (CPPN) [Stanley, 2007] is an efficient approach able to generate morphologies with symmetry and repetitive structures. In contrast, direct encoding often uses graph representations for robot morphologies [Wang *et al.*, 2019], allowing effective modeling of relationships between modules. Beyond single-task co-design, this paradigm has been extended to multi-task settings [Mertan and Cheney, 2024b], facilitating knowledge transfer from simple tasks to more complex ones [Zhao *et al.*, 2025b; Wang *et al.*, 2023d]. In addition, generative models were incorporated into co-design, including classifier-guided diffusion models to improve the efficiency of morphological evolution [Wang *et al.*, 2023a; Liu *et al.*, 2025c]. Large language models have also been explored as evolutionary search operators to balance optimization efficiency with solution diversity [Song *et al.*, 2025; Fang *et al.*, 2025].



(a) (b)

Figure 1: Example of a VSR. (a) Robot structure. (b) Corresponding digital encoding.

2.2 Self-Attention

Self-Attention has demonstrated significant potential in recent years across deep learning domains, particularly in Natural Language Processing (NLP) and Computer Vision (CV) [Tsai *et al.*, 2019; Wu *et al.*, 2022; Zhao *et al.*, 2025a; Parisotto *et al.*, 2020]. Unlike in traditional sequence models based on Recurrent Neural Networks (RNNs) or Convolutional Neural Networks (CNNs), the Transformer relies entirely on attention mechanisms [Vaswani *et al.*, 2017], eliminating the need for recurrent or convolutional structures. It learns attention relationships among all positions in the input sequence. The self-attention mechanism projects the input sequence through trainable weight matrices into query, key, and value vectors, defined as $Q = XW_Q, K = XW_K, V = XW_V$. Subsequently, it computes the similarity between queries and keys to obtain attention weights via $\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)$. These weights represent the relevance of each key position to each query position in the sequence. In our method, attention weights are used to assign importance values to robot components, enabling the model to adaptively focus on structural elements that critically impact task performance during decision-making. This improves the overall representation capability and optimization efficiency.

3 Problem Formulation

Voxel-based Soft Robots (VSRs) are a class of deformable robotic structures composed of discrete, regularly arranged voxel units. Each voxel typically corresponds to an elastic material unit capable of local extension or compression. Through mechanical coupling and force transmission between adjacent voxels, VSRs achieve holistic shape evolution and motion generation. Compared with continuum soft robots, VSRs are modular and facilitate modeling, simulation, and optimization. Hiller et al. [2011] first introduced the concept of voxel-based soft robots. Subsequent work demonstrated diversity in morphology and behavior, suggesting that voxelization is an effective approach for studying robot co-design [Cheney *et al.*, 2014]. In Figure 1, the 2D VSR used in this study consists of five types of voxels: empty, hard, soft,

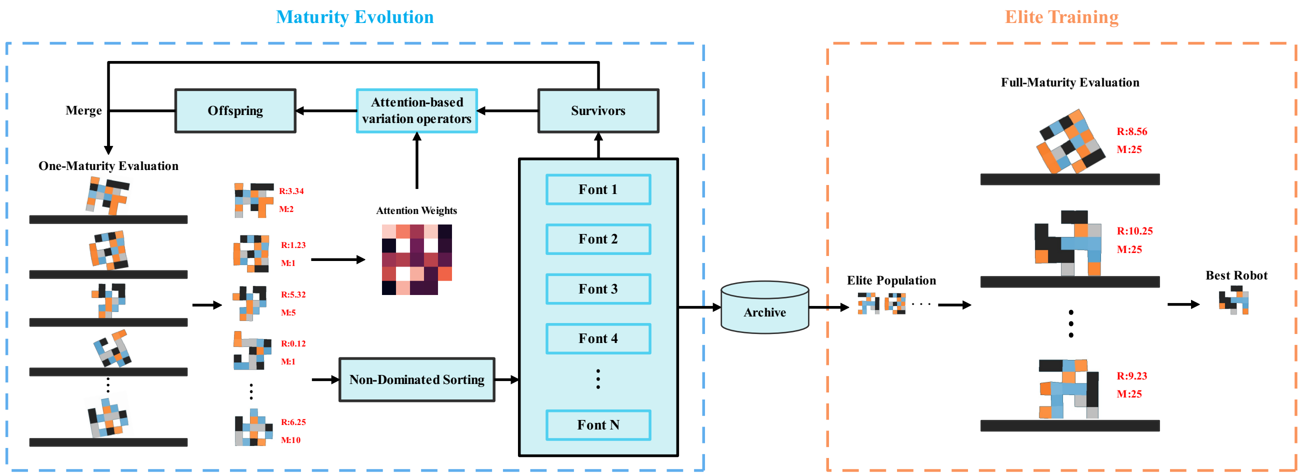


Figure 2: Overview of AGCD. It consists of two phases: the Maturity Evolution Phase, which evolves high-quality individuals using Maturity–Fitness Non-Dominated Sorting and an Attention-Based Variation Operator; and the Elite Training Phase, which conducts high-maturity evaluation for all low-maturity, high-potential individuals collected in the previous phase.

horizontal actuator, and vertical actuator, encoded as the digits 0, 1, 2, 3, and 4 respectively. Only the horizontal and vertical actuator voxels can actively deform, producing strain proportional to the original length along the corresponding direction. Soft voxels undergo passive deformation only, whereas rigid voxels do not deform. For further details on this simulation platform, refer to [Bhatia *et al.*, 2021].

The co-design problem for VSRs is to search, within a given search space, for the optimal voxel-robot morphology for the target task and for the optimal controller for this morphology. This problem can be formulated as a bi-level optimization problem:

$$\begin{aligned}(s^*, \phi_s^*) &= \arg\max_{s \in \mathcal{S}} \; F(s, \phi_s^*)\,, \\ \text{s.t. } \phi_s^* &= \arg\max_{\phi \in \Theta} \; f(s, \phi)\,.\end{aligned} \tag{1}$$

where s is the robot morphology and ϕ is the corresponding controller, which is typically represented by a neural network. $\mathcal{S}$ is the morphology search space, which is vast and discrete. In EvoGym, for an $n \times n$ robot, the size of this space is 5^{n^2} when constraints are not considered. $F(\cdot)$ is the outer-level objective that aims to find the optimal morphology s^*. $f(\cdot)$ is the inner-level objective that solves for the optimal control policy ϕ_s^* for a given morphology. This inner-level optimization is typically solved using reinforcement learning (RL), but it requires collecting a large amount of simulation data to update network parameters, making the process computationally expensive. One of the core objectives of this work is to enhance the efficiency of computational resource allocation, allowing costly controller training to focus primarily on potentially high-quality morphologies, thereby improving overall optimization efficiency.

4 Proposed Method

In this work, we propose AGCD, which applies attention guidance to both the design and control phases. Figure 2 presents the overall framework of this method, which consists of a Maturity Evolution and an Elite Training. In the first phase, we employed low-maturity evaluation. The evaluated population is used to update the robot archive and to extract attention weights. We then conduct stratified ranking via Maturity-Fitness Non-Dominated Sorting and retain the top-ranked individuals. Offspring are generated using the attention-based variation operator, and this procedure is repeated in each generation. In the second phase, low-maturity individuals with high potential are selected from the robot archive and trained to high maturity. The comprehensive procedure is provided in Algorithm 1. The details of multi-maturity evaluation and Maturity-Fitness Non-Dominated Sorting are introduced in Sections 4.1 and 4.2, respectively. The attention-based variation operator are discussed in Section 4.3. To improve control performance, a self-attention control strategy is adopted, and coordinate-based positional embeddings are used to enhance the positional information in the observations, outlined in Section 4.4.

4.1 Multi-Maturity Evaluation

To balance exploration efficiency and evaluation reliability with a limited computational budgets, our work employs multi-maturity evaluation: a method incorporating different maturity levels, specifically low-maturity and high-maturity evaluations. Low-maturity evaluation trains controllers with a small, fixed number of iterations, the maturity level is then incremented by one. High-maturity evaluation fully trains controllers and sets the maturity level to its maximum value. Low-maturity evaluation is performed at each generation during the maturity evolution phase, while high-maturity evaluation is applied at the elite training phase. In Appendix E, we experimentally demonstrated that setting the maximum maturity to 25 yielded optimal results. Given the above evaluation approach, each individual has two evaluation metrics: fitness and maturity. To better utilize these metrics, we introduce the Maturity-Fitness Non-dominated Sorting. The specific dominance rules and sorting method are as follows:

4.2 Maturity-Fitness Non-Dominated Sorting

Non-dominated sorting is a widely used ranking method for multi-objective optimization [Saito *et al.*, 2022; Srinivas and Deb, 1994; Ma *et al.*, 2023]. Its core idea is to define a dominance relation that hierarchically organizes candidate solutions. This work employs the Pareto dominance rule, where maturity is minimized and fitness is maximized. The Maturity-Fitness Pareto dominance rule is defined as follows:

$$a \text{ dominates } b = \begin{cases} a_m \leq b_m, \\ a_f \geq b_f, \\ (a_m < b_m) \vee (a_f > b_f) \end{cases} \tag{2}$$

where a_m and b_m are the maturity and a_f and b_f the fitness of individual a and individual b, respectively. Individual a is Pareto-superior to individual b if and only if the following conditions are satisfied: a is no greater than b in maturity; a is no smaller than b in fitness; and a is strictly better than b in at least one objective. In other words, this relation requires that individual a is not inferior to b in all objectives and only strictly superior to b in at least one objective.

In our implementation of non-dominated sorting, the population is divided into several frontier layers ranked from best to worst, with the first frontier layer consisting of individuals not dominated by any other individual. The sorting process follows the NSGA-II fast non-dominated sorting strategy [Srinivas and Deb, 1994], as detailed in Algorithm 2. Within the same frontier layer, individuals are sorted by fitness. Under the above rules, Maturity-Fitness Non-Dominated Sorting protects immature individuals and allocates computation to high-potential individuals.

4.3 Attention-Based Variation Operator

We observe that the controller assigns different attention weights to voxels at different positions. Higher attention indicates greater importance of that position for the current control decision. Following each individual evaluation, we run a simulation rollout using the corresponding trained controller and extracts the attention matrices at each time step. Subsequently, the attention matrices are averaged across time steps and normalized to obtain an attention weights for each voxel of the individual. This score is then used to guide subsequent crossover and mutation operations.

Attention-Based Crossover Operation

Attention-Guided Crossover Operation treats attention weights as a structural bias to guide the evolution, increasing the probability that offspring inherit high-contribution traits.Let v_i^c and a_i^c are the numerically encoded voxel type and its corresponding attention weights for parent P_i ($i \in \{1, 2\}$) at position c. The offspring's voxel at each

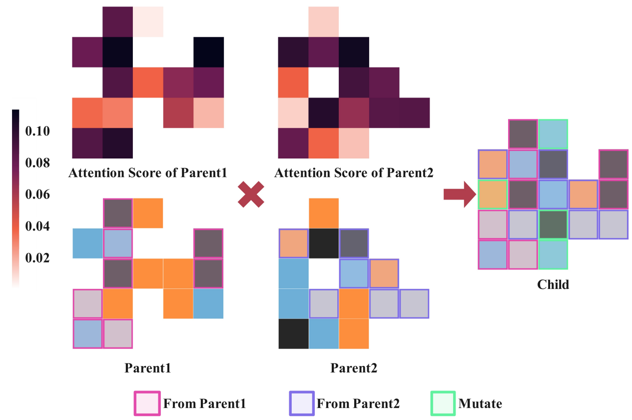


Figure 3: Example of the Attention-Based Crossover Operation.

position is updated according to the following rule:

$$v_{off}^c = \begin{cases} v^c_{\arg\max_i a_i^c} & \text{if } v_1^c, v_2^c \neq 0 \\ v_i^c & \text{if only } v_i^c \neq 0, a_i^c \geq \tilde{a}_i \\ mutate(c) & \text{if only } v_i^c \neq 0, a_i^c < \tilde{a}_i \\ 0 & \text{if } v_1^c = v_2^c = 0 \end{cases} \tag{3}$$

where $\tilde{a}_i$ is the median attention weights of all non-empty voxels in parent P_i, and $mutate(c)$ is a random mutation that changes the voxel at position c to another voxel type. After offspring generation, offspring undergo structural feasibility checks. If the requirements are not satisfied, the process is repeated until the predefined upper limit is reached. Figure 3 illustrates an example of this operation, where pink borders represent voxels inherited by the offspring from Parent1, purple borders represent voxels inherited from Parent2, and green borders denote newly mutated voxels.

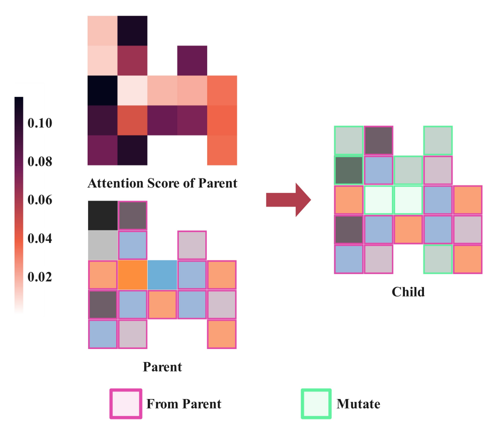


Figure 4: Example of the Attention-Based Mutation Operation.

Attention-Based Mutation Operation

The attention-based mutation operation is similar to the above crossover operation, taking a single parent structure and its

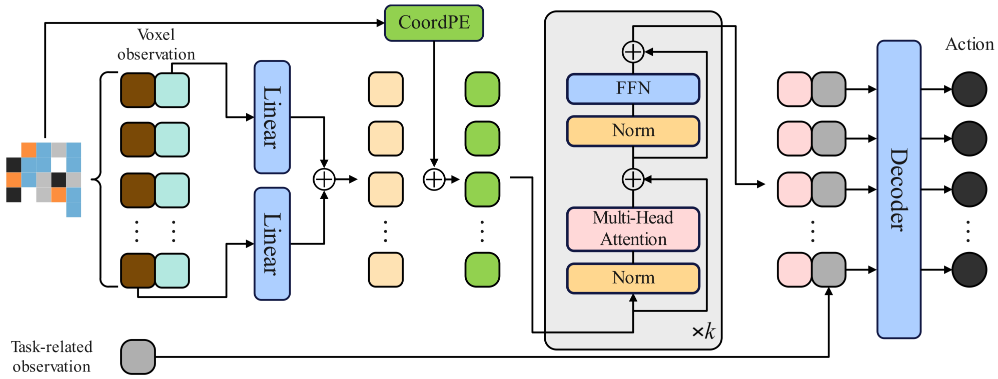


Figure 5: Control strategy based on a self-attention architecture. First, two linear layers encode the voxel observations in two directions. Then, coordinate-based positional embeddings enhance positional information. Finally, a Transformer models global dependencies to achieve an efficient control strategy.

associated attention weights as input. Let v^c and a^c denote the numerically encoded voxel type and its corresponding attention weight at position c in parent P. The voxel of the offspring at each position is updated according to the following rule:

$$v_{off}^c = \begin{cases} v^c & \text{if } v^c \neq 0 \text{ and } a^c \geq \tilde{a} \\ mutate(c) & \text{if } v^c \neq 0 \text{ and } a^c < \tilde{a} \\ mutate(c) & \text{if } v^c = 0 \end{cases} \tag{4}$$

where $\tilde{a}$ is the median attention weight of all non-empty voxels in parent P, and $mutate(c)$ is a random mutation that changes the voxel at position c to another voxel type. After offspring generation, structural feasibility checks are performed. If requirements are not met, the process retries until reaching the predefined upper limit. Figure 4 illustrates an example of this operation, depicting that regions with high attention weights are preserved while low-scoring areas are mutated into different voxels.

Both of these variation operators simultaneously avoid damaging critical structures while inducing controlled perturbations in low-contribution regions, thereby enhancing both the diversity of structural exploration and overall evolutionary efficiency.

4.4 Control and Training Strategies

For the robot to explore regions with varying importance at different positions, we employ a Transformer architecture to design a control strategy based on self-attention. Figure 5 illustrates the network architecture of the control strategy. At each time step t, the observation $s_t^k = \{s_t^l, s_t^r\}$ serves as input for the controller. The observation s_t^l is the voxel observation, specifically the relative position coordinates of the four corner points and the center point of each voxel. The observation s_t^r is the task-related observation, such as velocity and target position. The observation s_t^l includes information along the x and y axes. To better process information in these two directions, two linear layers are used to embed the observations $s_{t,x}^k$ and $s_{t,y}^k$ separately, and the resulting embeddings are concatenated to form the observation embedding e_t^k. The specific representation is as follows:

$$e_t^k = \left[f\left(s_{t,x}^k\right); f\left(s_{t,y}^k\right) \right] \tag{5}$$

where $f(.)$ is the linear layer and $[.]$ is concatenation. To enhance positional information, we employ coordinate-based positional embeddings (CoordPE), with $p_x^{k^i}, p_y^{k^i} \in \mathbb{N}$ representing the x-coordinate and y-coordinate of voxel k^i, respectively. We employ learnable embedding vectors $\mathbf{p}_x^{k^i}, \mathbf{p}_y^{k^i}$ to represent the positional information of each voxel k^i. Specifically:

$$\begin{aligned} \mathbf{p}^{k^i} &= \left[\mathbf{p}_x^{k^i}; \mathbf{p}_y^{k^i} \right], \\ x_t^{k^i} &= e_t^{k^i} + \mathbf{p}^{k^i} \end{aligned} \tag{6}$$

where $x_t^{k^i}$ is the transformer input embedding for each voxel. The entire control policy can be expressed as $\pi_{\theta_K}\left(a_t^k | s_t^k, \Psi_t\right)$, where a_t^k is constrained to the range [0.6,1.6]. During training, the decoder outputs an action mean μ, which is used as the mean of a Gaussian distribution with a fixed diagonal covariance matrix. All actions are sampled from this Gaussian distribution during training, while the action mean is directly used during evaluation.

To obtain the sharing of control strategies, each newly generated individual inherits the controller from the individual in the current population whose morphology is most similar. Morphological similarity is measured by the sum of absolute differences between their digitally encoded voxel matrices, with a smaller sum indicating higher similarity. To train the control policy, we employ Proximal Policy Optimization (PPO) [Schulman *et al.*, 2017] as the core reinforcement learning framework. PPO enhances the stability of the optimization process by limiting the magnitude of updates between old and new policies.

5 Experiments

We evaluated whether our method could evolve high-quality robots by addressing the following questions: (1) Can attention effectively guide the search direction in co-design? (2)

How do the various components of AGCD respectively influence co-design? (3) Can our method perform well on unseen co-design tasks?

5.1 Experimental Settings

All experiments were implemented on the Evolution Gym. The maximum iteration count per individual for each task is referenced in Table 4. We set the design space of the robot to 5x5, defined the maximum training iterations t per individual for each task, and set the total iterations to $100t$ for easy tasks, $200t$ for medium tasks, and $300t$ for hard tasks. To comprehensively evaluate the performance of AGCD, we validated our method on 27 task environments, comprising 9 easy tasks, 9 medium tasks, and 9 hard tasks. These tasks include Walking, Object Manipulation, Climbing, and Locomotion. To better compare differences with other methods, we designed 3 baseline categories as follows:

- **GA**: Genetic Algorithm is a classic evolutionary computation method that simulates natural selection, inheritance, and mutation mechanisms to select and mutate candidate solutions, iteratively generating superior individuals.
- **BO**: Bayesian Optimization is a surrogate-assisted algorithm used to solve expensive black-box optimization problems [Wang *et al.*, 2023b]. It typically employs Gaussian processes to approximate the true objective function, thereby reducing computational costs associated with expensive fitness evaluations.
- **AIEA**: Act_Inh-based EA is a fast fitness evaluation algorithm [Liu *et al.*, 2023]. Candidate robots inherit actions generated by well-trained controllers, combine them with error estimation, and obtain approximate task performance.

AGCD adopts the $(\mu + \lambda)$ evolutionary strategy [Beyer and Schwefel, 2002], where $\mu = 10$ and $\lambda = 10$. To maintain consistency across experiments, the population size for all other methods was set to 20, and PPO parameter settings remained identical. For details, please refer to Appendix B.

5.2 Results and Analysis

Task	AGCD(Ours)	AIEA	GA	BO
Walker-v0	***10.52±0.25**	*10.35±0.43	9.40±0.87	7.58±0.39
Flipper-v0	**5.09±0.30**	4.67±0.23	4.09±0.68	2.76±0.53
Pusher-v1	**0.60±0.02**	0.19±0.01	0.16±0.04	0.10±0.01
Thrower-v0	**2.21±0.16**	0.94±0.06	0.83±0.06	0.84±0.02
Hurdler-v0	**3.05±0.21**	1.49±0.10	1.28±0.08	1.04±0.25
GapJumper-v0	**8.26±0.44**	3.53±0.19	2.21±0.31	3.08±0.37

Table 1: Mean and standard error of all results across selected tasks. Best results are highlighted in bold.* indicates that the result reached the upper bound.

In Figure 6 and Table 1, we present the reward curves and final mean values of AGCD, AIEA, GA, and BO on selected representative tasks. The results indicate that on easy tasks, our method performed slightly better than the other methods, with small performance differences. However, for medium and hard tasks, our method clearly achieved a superior performance compared to the other methods. (For space consideration, the reward curves and final fitness values for all tasks are provided in Appendix D). This phenomenon mainly results from the low challenge of easy tasks, which makes it harder to observe differences between methods. However, on medium and hard tasks, our approach showed an evident advantage, suggesting that attention can effectively guide the search process in co-design. In co-design, a larger design space search volume increases the likelihood of discovering high-quality individuals. Notably, our method explores a larger search volume than other approaches. As shown in Appendix Table 6, AGCD achieved the largest average search volume on all tasks, followed by AIEA, and both are significantly larger than those of GA and BO as AGCD and AIEA include individuals with partial or approximate evaluation, while GA and BO require full evaluation for every individual. From this perspective, AGCD achieves higher search efficiency. An-

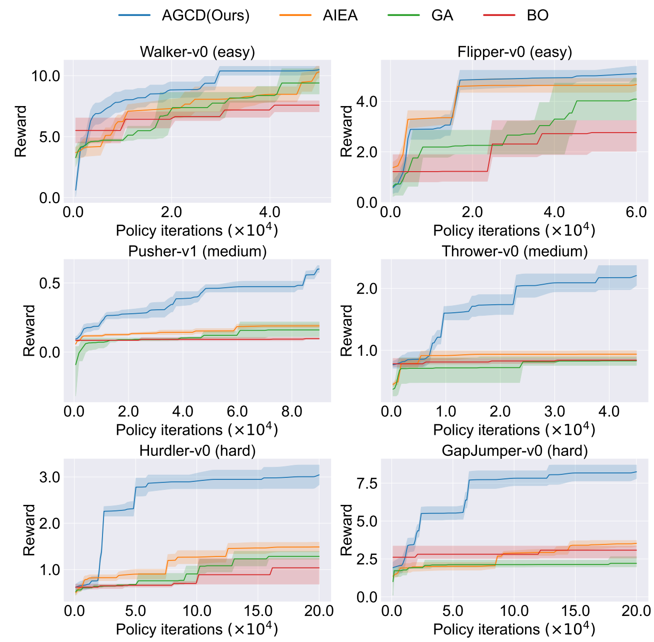


Figure 6: Mean and standard deviation of reward curves for selected tasks with AGCD, AIEA, GA, and BO. Each curve is averaged over five independent runs.

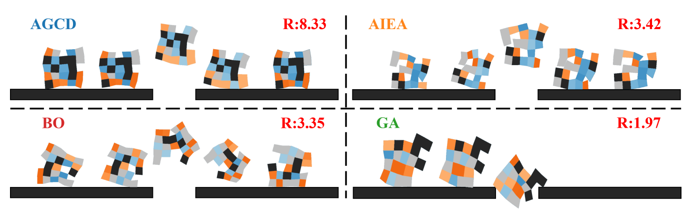


Figure 7: Visualization results of AGCD, AIEA, GA, and BO on the GapJumper-v0 task. The figure shows changes in robot position and motion over time for each method. Snapshots are arranged from left to right in chronological order. The evolved AGCD robot demonstrated crab-like lateral running with the ability to jump across gaps.

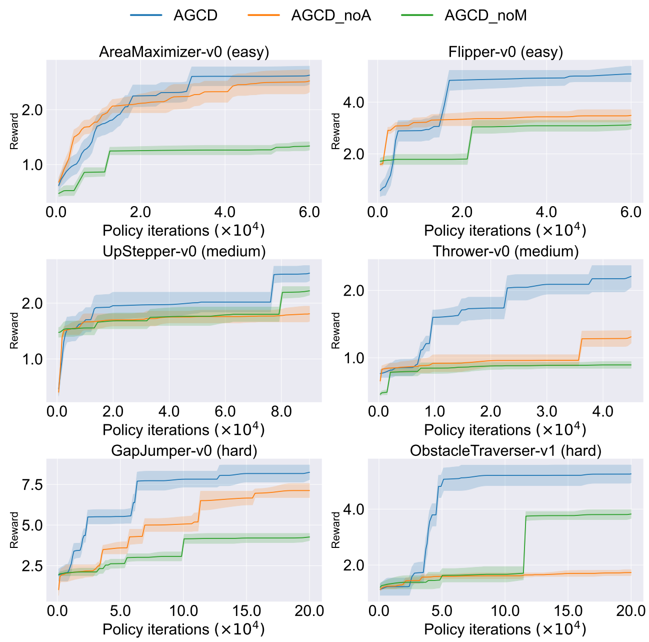


Figure 8: Mean and standard deviation of reward curves for selected tasks with AGCD, AGCD-noA and AGCD-noM. Each curve is averaged over five independent runs.

other reason is that AGCD has higher efficiency in computational resource allocation. Appendix F showes the fitness distributions on selected tasks. The fitness value at the density peak for our method was clearly higher than those of the other baseline methods, and the overall distribution shifted toward higher fitness ranges.

Figure 7 depicts visually the performance of all methods on the hard task GapJumper-v0. This task's objective is to jump across the gaps between platforms, with the reward increasng as the robot traverses more platforms. The results show that AGCD generated robots with crab-like running and jumping capabilities. In contrast, robots evolved using AIEA and BO exhibited unstable jumping behaviors, while those evolved using GA were obstructed at the gaps and failed to learn effective jumping.

5.3 Ablation Study

To validate the importance of each component of AGCD, we constructed two AGCD variants with all other parameters kept unchanged. The two variants are defined as follows:

- **AGCD-noA**: AGCD without its attention-based variation operator and replaced with the variation operator used in the other baseline algorithms.
- **AGCD-noM**: AGCD with multi-maturity evaluation and Maturity-Fitness Non-Dominated Sorting are removed. High-maturity evaluation and fitness-based are adopted but maturity is not considered.

Figure 8 shows the ablation study results on selected tasks. AGCD outperformed both AGCD-noA and AGCD-noM. On all tasks, removing a single component caused a sharp drop in performance, which supporting the importance of the

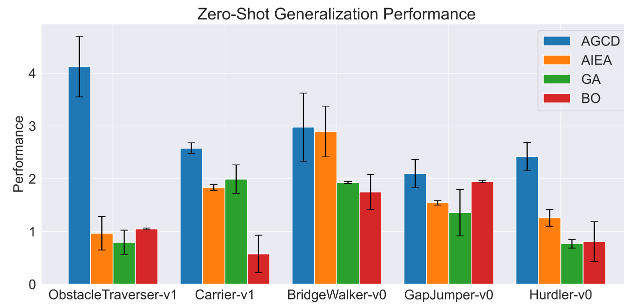


Figure 9: Evaluation results for zero-shot generalization. Bar height denotes average zero-shot performance, and error bars indicate the standard deviation.

attention-based variation operator and non-dominated maturity fitness sorting in AGCD. The fact that AGCD outperformed AGCD-noA indicates that the attention-based variation operator can better focus on critical structural regions, thereby generating higher-quality offspring with greater potential, suggesting in turn that attention can guide the evolutionary search toward higher quality candidate solutions. In addition, AGCD outperformed AGCD-noM suggesting that removing Maturity-Fitness Non-Dominated Sorting weakens the directionality and stability of the search process, thereby increasing the risk of falling into local optima and that multi-maturity evaluation allows effective choosing of allocation of computational resources.

5.4 Zero-Shot Generalization

To test whether our method can generate effective robots for unseen tasks, we compared zero-shot generalization performance across our method and all baseline approaches. A total of 5 pairs of basic tasks and target tasks are defined as follows: (1) More rugged terrain: from ObstacleTraverser-v0 to ObstacleTraverser-v1 (2) Larger obstacles: from ObstacleTraverser-v0 to Hurdler-v0 (3) Softer terrain: from Walker-v0 to BridgeWalker-v0 (4) Added box-placement task: from Carrier-v0 to Carrier-v1 (5) Different but equally difficult tasks: from PlatformJumper-v0 to GapJumper-v0. As shown in Figure 9, robots generated by our method on basic tasks performed well on target tasks, outperforming other baseline methods.

6 Conclusion

In this work, inspired by the Baldwin effect, we propose AGCD, an attention-guided co-design approach intended to generate high-quality robots and optimal controllers using a limited computational budget. Experimental results demonstrated that this method outperforms other baselines in search efficiency, final performance, and zero-shot generalization. Moreover, each module of AGCD is indispensable for these improvements. Overall, AGCD provides a feasible and effective strategy for improving performance on complex optimization problems with efficient use of computational resources and will serve as a useful foundation for future research.

References

[Beyer and Schwefel, 2002] Hans-Georg Beyer and Hans-Paul Schwefel. Evolution strategies–a comprehensive introduction. *Natural computing*, 1(1):3–52, 2002.

[Bhatia *et al.*, 2021] Jagdeep Bhatia, Holly Jackson, Yunsheng Tian, Jie Xu, and Wojciech Matusik. Evolution gym: A large-scale benchmark for evolving soft robots. *Advances in Neural Information Processing Systems*, 34:2201–2214, 2021.

[Bongard and Pfeifer, 2003] Josh C Bongard and Rolf Pfeifer. Evolving complete agents using artificial ontogeny. In *Morpho-functional machines: the new species: designing embodied intelligence*, pages 237–258. Springer, 2003.

[Cheney *et al.*, 2014] Nick Cheney, Robert MacCurdy, Jeff Clune, and Hod Lipson. Unshackling evolution: evolving soft robots with multiple materials and a powerful generative encoding. *SIGEVOlution*, 7(1):11–23, August 2014.

[Cheney *et al.*, 2016] Nicholas Cheney, Josh Bongard, Vytas SunSpiral, and Hod Lipson. On the difficulty of co-optimizing morphology and control in evolved virtual creatures. In *Artificial life conference proceedings*, pages 226–233. MIT Press One Rogers Street, Cambridge, MA 02142-1209, USA journals-info . . . , 2016.

[Cheney *et al.*, 2018] Nick Cheney, Josh Bongard, Vytas SunSpiral, and Hod Lipson. Scalable co-optimization of morphology and control in embodied machines. *Journal of The Royal Society Interface*, 15(143):20170937, 2018.

[Fang *et al.*, 2025] Jiawei Fang, Yuxuan Sun, Chengtian Ma, Qiuyu Lu, and Lining Yao. Robomore: Llm-based robot co-design via joint optimization of morphology and reward. *arXiv preprint arXiv:2506.00276*, 2025.

[Harada and Iba, 2024] Kazuaki Harada and Hitoshi Iba. Lamarckian co-design of soft robots via transfer learning. In *Proceedings of the Genetic and Evolutionary Computation Conference*, pages 832–840, 2024.

[Hiller and Lipson, 2011] Jonathan Hiller and Hod Lipson. Automatic design and manufacture of soft robots. *IEEE Transactions on Robotics*, 28(2):457–466, 2011.

[Lipson and Pollack, 2000] Hod Lipson and Jordan B Pollack. Automatic design and manufacture of robotic lifeforms. *Nature*, 406(6799):974–978, 2000.

[Liu *et al.*, 2023] Shulei Liu, Wen Yao, Handing Wang, Wei Peng, and Yang Yang. Rapidly evolving soft robots via action inheritance. *IEEE Transactions on Evolutionary Computation*, 2023.

[Liu *et al.*, 2025a] Guiliang Liu, Bo Yue, Yi Jin Kim, and Kui Jia. Embracing evolution: A call for body-control co-design in embodied humanoid robot. *arXiv preprint arXiv:2510.03081*, 2025.

[Liu *et al.*, 2025b] Huaping Liu, Di Guo, and Angelo Cangelosi. Embodied intelligence: A synergy of morphology, action, perception and learning. *ACM Computing Surveys*, 57(7):1–36, 2025.

[Liu *et al.*, 2025c] Shulei Liu, Junchi Yan, Handing Wang, and Yaochu Jin. Morphology evolution for embodied robot design with a classifier-guided diffusion model. *IEEE Transactions on Evolutionary Computation*, 2025.

[Lu *et al.*, 2025] Haofei Lu, Zhe Wu, Junliang Xing, Jianshu Li, Ruoyu Li, Zhe Li, and Yuanchun Shi. Bodygen: Advancing towards efficient embodiment co-design. In *The Thirteenth International Conference on Learning Representations*, 2025.

[Ma *et al.*, 2023] Haiping Ma, Yajing Zhang, Shengyi Sun, Ting Liu, and Yu Shan. A comprehensive survey on nsga-ii for multi-objective optimization and applications. *Artificial Intelligence Review*, 56(12):15217–15270, 2023.

[Mertan and Cheney, 2024a] Alican Mertan and Nick Cheney. Investigating premature convergence in co-optimization of morphology and control in evolved virtual soft robots. In *European Conference on Genetic Programming (Part of EvoStar)*, pages 38–55. Springer, 2024.

[Mertan and Cheney, 2024b] Alican Mertan and Nick Cheney. Towards multi-morphology controllers with diversity and knowledge distillation. In *Proceedings of the Genetic and Evolutionary Computation Conference*, pages 367–376, 2024.

[Mertan and Cheney, 2025] Alican Mertan and Nick Cheney. Evolutionary brain-body co-optimization consistently fails to select for morphological potential. In *Artificial Life Conference Proceedings 37*, volume 2025, page 63. MIT Press One Rogers Street, Cambridge, MA 02142-1209, USA journals-info . . . , 2025.

[Parisotto *et al.*, 2020] Emilio Parisotto, Francis Song, Jack Rae, Razvan Pascanu, Caglar Gulcehre, Siddhant Jayakumar, Max Jaderberg, Raphaël Lopez Kaufman, Aidan Clark, Seb Noury, Matthew Botvinick, Nicolas Heess, and Raia Hadsell. Stabilizing transformers for reinforcement learning. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7487–7498. PMLR, 13–18 Jul 2020.

[Pfeifer *et al.*, 2006] Rolf Pfeifer, Fumiya Iida, and Gabriel Gómez. Morphological computation for adaptive behavior and cognition. In *International Congress Series*, volume 1291, pages 22–29. Elsevier, 2006.

[Pigozzi *et al.*, 2023] Federico Pigozzi, Federico Julian Camerota Verdù, and Eric Medvet. How the morphology encoding influences the learning ability in body-brain co-optimization. In *Proceedings of the Genetic and Evolutionary Computation Conference*, pages 1045–1054, 2023.

[Rosendo *et al.*, 2017] Andre Rosendo, Marco Von Atzigen, and Fumiya Iida. The trade-off between morphology and control in the co-optimized design of robots. *PloS one*, 12(10):e0186107, 2017.

[Saito *et al.*, 2022] Takumi Saito, Haruki Nishimura, and Mizuki Oka. Comparative studies of evolutionary methods and rl for learning behavior of virtual creatures. In *2022 IEEE Symposium Series on Computational Intelligence (SSCI)*, pages 1059–1065. IEEE, 2022.

[Schulman *et al.*, 2017] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[Sims, 1994] Karl Sims. Evolving virtual creatures. In *Proceedings of the 21st Annual Conference on Computer Graphics and Interactive Techniques*, SIGGRAPH '94, page 15–22, New York, NY, USA, 1994. Association for Computing Machinery.

[Song *et al.*, 2025] Junru Song, Yang Yang, Huan Xiao, Wei Peng, Wen Yao, and Feifei Wang. Laser: Towards diversified and generalizable robot design with large language models. In *The Thirteenth International Conference on Learning Representations*, 2025.

[Srinivas and Deb, 1994] Nidamarthi Srinivas and Kalyanmoy Deb. Muiltiobjective optimization using nondominated sorting in genetic algorithms. *Evolutionary computation*, 2(3):221–248, 1994.

[Stanley *et al.*, 2009] Kenneth O Stanley, David B D'Ambrosio, and Jason Gauci. A hypercube-based encoding for evolving large-scale neural networks. *Artificial life*, 15(2):185–212, 2009.

[Stanley, 2007] Kenneth O Stanley. Compositional pattern producing networks: A novel abstraction of development. *Genetic programming and evolvable machines*, 8(2):131–162, 2007.

[Tanaka and Aranha, 2022] Fabio Tanaka and Claus Aranha. Co-evolving morphology and control of soft robots using a single genome. In *2022 IEEE Symposium Series on Computational Intelligence (SSCI)*, pages 1235–1242. IEEE, 2022.

[Tsai *et al.*, 2019] Yao-Hung Hubert Tsai, Shaojie Bai, Makoto Yamada, Louis-Philippe Morency, and Ruslan Salakhutdinov. Transformer dissection: a unified understanding of transformer's attention via the lens of kernel. *arXiv preprint arXiv:1908.11775*, 2019.

[Vaswani *et al.*, 2017] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[Wang *et al.*, 2019] Tingwu Wang, Yuhao Zhou, Sanja Fidler, and Jimmy Ba. Neural graph evolution: Towards efficient automatic robot design. *arXiv preprint arXiv:1906.05370*, 2019.

[Wang *et al.*, 2023a] Tsun-Hsuan Johnson Wang, Juntian Zheng, Pingchuan Ma, Yilun Du, Byungchul Kim, Andrew Spielberg, Josh Tenenbaum, Chuang Gan, and Daniela Rus. Diffusebot: Breeding soft robots with physics-augmented generative diffusion models. *Advances in Neural Information Processing Systems*, 36:44398–44423, 2023.

[Wang *et al.*, 2023b] Xilu Wang, Yaochu Jin, Sebastian Schmitt, and Markus Olhofer. Recent advances in bayesian optimization. *ACM Computing Surveys*, 55(13s):1–36, 2023.

[Wang *et al.*, 2023c] Yuxing Wang, Shuang Wu, Haobo Fu, Qiang Fu, Tiantian Zhang, Yongzhe Chang, and Xueqian Wang. Curriculum-based co-design of morphology and control of voxel-based soft robots. In *The Eleventh International Conference on Learning Representations*, 2023.

[Wang *et al.*, 2023d] Yuxing Wang, Shuang Wu, Tiantian Zhang, Yongzhe Chang, Haobo Fu, Qiang Fu, and Xueqian Wang. Preco: Enhancing generalization in co-design of modular soft robots via brain-body pre-training. In *Conference on Robot Learning*, pages 478–498. PMLR, 2023.

[Wu *et al.*, 2022] Leijie Wu, Song Guo, Yaohong Ding, Junxiao Wang, Wenchao Xu, Richard Yida Xu, and Jie Zhang. Demystify self-attention in vision transformers from a semantic perspective: analysis and application. *arXiv preprint arXiv:2211.08543*, 2022.

[Xu *et al.*, 2021] Jie Xu, Tao Chen, Lara Zlokapa, Michael Foshey, Wojciech Matusik, Shinjiro Sueda, and Pulkit Agrawal. An end-to-end differentiable framework for contact-aware robot design. *arXiv preprint arXiv:2107.07501*, 2021.

[Zhao *et al.*, 2024] Jiliang Zhao, Wei Peng, Handing Wang, Wen Yao, and Weien Zhou. A morphological transfer-based multi-fidelity evolutionary algorithm for soft robot design. *IEEE Computational Intelligence Magazine*, 19(4):16–30, 2024.

[Zhao *et al.*, 2025a] Jiliang Zhao, Wei Peng, Handing Wang, and Wen Yao. A multi-objective optimization framework based on information sharing for serially connected robot design. *Complex & Intelligent Systems*, 11:410, 2025.

[Zhao *et al.*, 2025b] Jiliang Zhao, Wei Peng, Handing Wang, Weien Zhou, Yang Yang, and Wen Yao. Cross-task collaborative optimization based on knowledge transfer for soft robot design. *IEEE Transactions on Evolutionary Computation*, 2025.